

Smartphone Education: Building a Modular Student-Designed Smartphone

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A Project Proposal for the Learn Through Discovery Projects Hatchery at California State Polytechnic University Pomona

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Project Summary

The Smartphone Project reimagines mobile technology by addressing key issues in user control, sustainability, and accessibility. In a world dominated by proprietary systems and closed ecosystems, our initiative introduces a fully modular, repairable smartphone platform designed from scratch by students. Our target audience includes students, open-source advocates, and tech enthusiasts who seek greater ownership over their devices and want to challenge the norms of mainstream design.

At its core, this project is an educational tool — a means of lowering the barrier to entry for students interested in hardware-software integration, systems design, and mobile technology. Using field-programmable gate arrays (FPGAs), open-source tools like KiCad, and modular component architecture (e.g., RAM, SD, RF modules), students will engage directly in the engineering of a fully functioning smartphone. From processor development and thermal management to PCB layout and Linux-based OS integration, this hands-on experience provides exposure to real-world engineering challenges that are often inaccessible in traditional classroom settings.

The Smartphone Project fills a significant gap in current engineering education. While automotive programs like FSAE provide pipelines for student-led development in the automotive industry, no comparable structure exists for the computer sector — despite its prevalence in everyday life and industry. Our project aims to build that pipeline, offering students a unique opportunity to gain interdisciplinary experience in one of the world's most influential technologies.

Unlike existing open-source phone initiatives that often compromise on usability or accessibility, our focus is on creating a product that is both functional and open — a balance that allows us to educate and inspire a broader, more diverse student community. This approach promotes sustainability through upgradability and repairability, reduces e-waste, and fosters a culture of ethical, transparent design.

Ultimately, this project is more than a device. It's a platform for student-driven innovation — one that empowers students to rethink what technology can be and who gets to build it.

Project Background and Solution

Despite the smartphone's ubiquity in daily life and its central role in today's tech industry, hands-on smartphone development is virtually absent in engineering education nationwide. While many universities offer project-based programs in automotive, aerospace, and robotics engineering — such as Formula SAE and CubeSat teams — few, if any, student-led groups or academic programs tackle embedded mobile systems at the hardware level. This is a striking absence, considering how essential smartphones have become in global markets, innovation, and daily life.

Organizations like IEEE provide value through networking and smaller technical workshops, but their focus often leans toward communications and signal processing, rather than immersive, system-level design challenges like smartphone architecture or PCB integration (IEEE, 2024). Their projects also tend to be beginner-friendly and less interdisciplinary, lacking the depth required to meaningfully prepare students for real-world systems engineering.

This gap is part of a broader issue in engineering education: the lack of large-scale, real-world interdisciplinary learning opportunities. As Kang and Liang (2024) note in their analysis of electrical and electronic engineering instruction, “There is a pressing need to incorporate real-world industry cases and hands-on project experience to help students better apply knowledge” (p. 1). Without such frameworks, students rarely have the opportunity to engage in multi-subsystem design, collaborative development, or systems integration — the skills that drive innovation in today’s most advanced technologies.

Even at Cal Poly Pomona, a campus known for its “learn by doing” philosophy, the gap is visible. While clubs like BroncoSpace, UMBRA, and Liquid Rocket Lab offer exceptional experiences in aerospace and mechanical engineering, students in Computer Engineering (CE), Electronic Systems Engineering Technology (ESET), Electromechanical Systems Engineering Technology (EMSET), and Electrical Engineering (EE) have few avenues for applied electronics and hardware-software co-design. Together, these majors account for over one-third of the entire engineering student population (CPP Institutional Research, 2023).

The Smartphone Project directly addresses this gap by offering a replicable model for interdisciplinary learning, centered on one of the most influential consumer technologies of the 21st century. Through this Hatchery-supported initiative, students build an FPGA-based smartphone from scratch — including its SoC logic, power systems, thermal management, RF communication, and Linux-based OS integration. More than just an engineering challenge, the project serves as an educational platform where students across subfields collaborate, experiment, and learn from failure.

Beyond education, the project responds to a broader critique of modern smartphone design — namely, that closed ecosystems, soldered components, and anti-repair policies have made devices less transparent, less upgradable, and harder to understand. For instance, iFixit’s analysis of the iPhone 15 Pro Max highlights how even small repairability gains are undercut by restrictive part-pairing software, earning the device a provisional score of 4/10 for repairability (iFixit, 2023). This philosophy encourages waste and prevents users — and students — from learning how these systems actually work.

Our project stands in contrast not only because it will be designed as a response to industry shortcomings, but also as a proof-of-concept for a new kind of engineering education. Unlike the proprietary chipsets produced by Apple, Qualcomm, or Samsung, FPGAs offer a blank, reprogrammable canvas — capable of acting as a processor, accelerator, or custom logic unit, depending on how it is configured. This reconfigurability opens the door to truly modular

hardware, where components can be upgraded, replaced, or redefined in software without scrapping the entire device.

By tackling both the educational gap and the design limitations of modern smartphones, the Smartphone Project aims to offer a replicable, scalable model for interdisciplinary engineering and offer a blueprint for other campuses. One that is inclusive, forward-thinking, and capable of reshaping how students engage with one of the most important technologies of the 21st century.

Objectives, Timeline, and Goals

The primary goal of the Smartphone Project is to design and prototype a fully modular, reconfigurable smartphone built by students, using field-programmable gate arrays (FPGAs), open-source tools, and custom-built subsystems. This initiative serves as both an educational platform and a demonstration of engineering capability. Through interdisciplinary collaboration, our goal is not only to develop a functioning device, but to create a sustainable foundation for future students to continue work on advanced mobile systems.

Our progress will be structured around six core technical phases, with defined deliverables at each step. These are aligned with funding milestones to ensure accountability and progress visibility:

Project Phases

Phase 1: Initial Planning and Documentation

In this phase, all teams define the specifications of their subsystems. The SoC team outlines the logic for CPU, GPU, and memory hierarchy. Hardware develops component selections and PCB interface definitions. RF outlines DSP and wireless system goals. The Enclosure team begins early CAD layouts. The Systems team ensures that documentation (use cases, requirements, and diagrams) align across subsystems.

Deliverables: Consolidated technical specification documents and architecture diagrams.

Phase 2: Subsystem Design and Simulation

Each team produces detailed subsystem designs. The SoC team models their logic blocks using Verilog and runs testbenches. Hardware simulates PCB behavior for signal integrity and power management. RF simulates frequency behavior and DSP logic. Enclosure finalizes thermals and spatial constraints.

Deliverables: Simulated subsystem models, refined schematics, and updated documentation.

Phase 3: Collaboration and Prototyping

This is our first major funding milestone and includes early prototyping and cross-team integration. SoC logic is implemented on FPGA dev boards. Hardware fabricates test PCBs. Enclosure produces 3D-printed prototypes. RF integrates antennas with FPGA logic. Teams validate interconnects and begin integration testing.

Deliverables: First-generation hardware prototypes, component alignment confirmation, and integration planning document.

Phase 4: Driver and Software Development

With hardware prototypes in hand, this phase begins full firmware and software development. SoC teams write drivers for memory and GPU modules. RF and Enclosure teams work on runtime monitoring (signal and thermal). The Systems team builds interface logic and startup sequences.

Deliverables: Driver suite and diagnostic tools capable of validating each subsystem independently.

Phase 5: System Integration and Final Testing

This is our second funding milestone, where we combine all systems into a working prototype. SoC and Hardware teams validate end-to-end processing. RF tests full communications functionality. Enclosure is tested under thermal load. The Systems team runs full-stack performance, thermal, and RF tests.

Deliverables: Fully integrated prototype that performs at least one real-world task, such as basic imaging or ML inference.

Phase 6: Final Review, Documentation, and Handoff

This is our final funding milestone. All teams will finalize documentation, generate onboarding material, and ensure reproducibility of the work. The SoC team wraps HDL, Hardware finalizes PCBs, RF documents DSP, and the Systems team assembles technical reports and test logs.

Deliverables: A full documentation package, final presentation, and knowledge transfer materials for new members.

Throughout this project, we will use our general meetings, demos, and Git-based repositories to showcase progress and disseminate our work to the broader tech community. Our goal is to release open-source documentation and design files via GitHub, along with blog-style updates through our website and social media.

A project timeline in the form of a Gantt chart is attached below and outlines all major milestones, and overall timeline up through the project's final review in August while trying not to clutter with too much information because each team has up to 30 subtasks each.

SMART Goals

Specific

Our team wants to build a smartphone from scratch that is comfortable to hold, intuitive to use, powerful enough to maintain strong performance, efficient enough to endure a whole day of extensive use, speedy enough to feel snappy and responsive, and versatile enough to serve as a foundation for future innovations while maintaining a clear focus on delivering exceptional core functionality in our first release.

Measurements for Success

User Defined in a Poll: “Comfortability to Hold”, “Intuitive to Use”, and “Speedy Enough to Feel Snappy and Responsive”

“Powerful enough to maintain strong performance” - Clock Speed, Voltage Regulation Test, Bandwidth, Power-Efficiency (Performance Per Watt)

“Efficient enough to endure a whole day of extensive use” - “Screen On Time”, “Real World Testing”, “Battery Capacity (mAh)”, “Battery Discharge Rate (mA)”, “Energy Efficiency Ratio (EER)”, “Component Power Draw”

Attainability

We have the fundamental knowledge necessary to build a smartphone from scratch with the qualities we listed in our goals. With that said, we may run into more sophisticated and technical road bumps in our project. In such cases, we have multiple avenues to address the situation including using Hatchery advisors/mentors and CPP Faculty, Textbooks, and Github.

Relevant

Yes. The overarching goals of our organization is to make learning the process of building a smartphone more frictionless. That starts with us being able to build a smartphone from scratch and proving that we can do it ourselves.

Time-Bound

Concept & Planning (Nov - Jan): Define project scope, secure funding, and conduct requirements analysis for all subteams.

Design & Procurement (Jan - March): Perform trade studies, finalize material selection, and develop initial design concepts. Begin purchasing components.

Development & Prototyping (March - May): Each subteam develops and integrates their respective systems, followed by system-wide testing.

Testing & Refinement (May - June): Perform electrical, RF, software, and thermal testing. Address risks and finalize design.

Finalization & Review (June - July): Conduct product readiness review and final design verification (CDR).

Personnel / Team Members

Sunny Nguyen

Role: Chief Technical Officer (CTO)

Responsibilities: Oversees the entire project, ensuring milestones and deadlines are met. Coordinates teams, manages resources, and communicates with stakeholders. Offers technical support and leads meetings, progress tracking, and quality assurance.

Qualifications: Designed the Celesti-Call Phone. Led engineering teams in rocket projects and CubeSats. Experienced in KiCad PCB design.

Chris Zambrana

Role: Project Manager

Responsibilities: Oversees the entire project, ensuring milestones and deadlines are met. Coordinates teams, manages resources, and communicates with stakeholders. Offers technical support and leads meetings, progress tracking, and quality assurance.

Qualifications: Designed a SoC on a FPGA with a Memory Mapped Input-Output Subsystem and Video subsystem. Knowledge in Computer Architecture and Operating Systems and Experienced in Digital Design Using HDL

Russell Chung

Role: Business Coordinator and Engineering Manager

Responsibilities: Leads proposal writing, funding applications, and sponsorship outreach. Manages project planning, including Gantt chart creation, timeline tracking, and deliverable coordination. Oversees budgeting, resource allocation, and milestone progress. Supervises engineering activities across subteams (Hardware, SoC, RF, Enclosure, etc.) and works with Systems Engineers to ensure cross-team interface compatibility. Organizes and leads weekly lead meetings while maintaining alignment across disciplines. Supports all teams by developing and running curriculum programs to boost team knowledge and repackage technical material for practical use. Manages external communication, student onboarding, and internal accountability to keep the team focused and on track.

Qualifications: Lead Successful Project Teams (Robotics, Outreach Organization Activities, Baseball). Experienced in technical writing, budget management, team management, and scalable curriculum development for hardware-software systems.

Max Gross

Role: Hardware Designer

Responsibilities: Designs and develops physical components, schematics, component selection, and PCB layout. Ensures integration with other subsystems and meets performance specs.

Qualifications: Designed radiation-testing PCB used at Lawrence Berkeley National Lab. Developed microcontroller ethernet at Astranis Space Corp. Led SUAS obstacle avoidance system.

Hoanh Lam

Role: System Engineer Lead

Responsibilities: Oversees hardware/software integration. Coordinates architecture definition, system testing, and optimization across subsystems.

Qualifications: Led CubeSat team in BLADE program. Northrop Grumman system engineering collaborator.

Nathan Alquicira

Role: System Engineer

Responsibilities: Supports integration and architecture. Shadows team members and assists in debugging and cross-functional collaboration.

Qualifications: Mentored robotics team. Experience with Metrolink PTC hardware/software issue management.

Viren Kumar

Role: Enclosure Engineer

Responsibilities: Designs the phone casing for structure and thermal performance. Collaborates with hardware team for spatial integration.

Qualifications: Experience in CAD and thermal considerations.

Danny Tran

Role: Back-end Developer

Responsibilities: Develops server-side software, APIs, and backend infrastructure. Supports embedded integration and system-level software optimization.

Qualifications: Developed Gemini API chatbot. IT technician with hardware/software debugging experience.

Izaiah Ortiz

Role: Front-end Developer

Responsibilities: Designs UI/UX for the OS and applications. Ensures responsiveness, aesthetics, and functionality across devices.

Qualifications: Experience in user interface development and cross-team collaboration.

Andy Zhang

Role: System-on-a-Chip Lead

Responsibilities: Leads SoC team. Develops FPGA-based drivers and HDL IPs. Coordinates with RF and hardware teams for integration and testing.

Qualifications: 8 years programming. Experience in HDL, drivers, reverse engineering, TCP/IP, and digital logic

Budget and Justification

Milestone 1: Design & Component Acquisition — Total: \$2,432.50

- This milestone funds the foundational components required to initiate design and early prototyping across all subsystems. This includes the procurement of FPGA development boards (for SoC logic testing), SoC and memory chips, early RF modules, camera modules, and power system components. Passive components and connectors are essential to create functional subsystems, while tools like microphones and LCDs are necessary for developing the phone's interactive features.

A portion of this milestone also supports our first round of PCB manufacturing, which enables the Hardware and SoC teams to move beyond simulation and validate physical designs. Thermal materials (pads, spreaders) are included to prepare for future

integration and allow for realistic testing conditions even in early stages. These purchases ensure that all teams can begin work concurrently and avoid bottlenecks from shared hardware.

- Liquid Crystal Display (LCD) Display Module (Qty: 2) — \$51.00

→ *Explanation: Used to prototype and test the graphical interface, allowing for development and debugging of the user interface system.*

- Near Field Communication (NFC) Controller (Qty: 2) — \$102.00

→ *Explanation: Enables wireless communication capabilities such as tap-to-pay and data transfer, showing our modular extensibility.*

- System-on-Chips (SoC) (Qty: 2) — \$288.00

→ *Explanation: Core processing element for running basic logic, including CPU and GPU operations — critical to SoC development and integration.*

- Memory Chips (Qty: 2) — \$47.60

→ *Explanation: Provides volatile and non-volatile memory support for software execution and application data storage.*

- Radio Frequency (RF) Chip (Qty: 2) — \$11.90

→ *Explanation: Supports wireless communication simulation and prototyping, essential for long-range data transmission.*

- FPGA Development Board (Qty: 4) — \$572.00

→ *Explanation: Serves as the backbone for logic development, including SoC prototyping and GPU compute core design.*

- Battery (Qty: 2) — \$51.00

→ *Explanation: Provides power delivery testing for the phone during prototyping, enabling full-device integration validation.*

- Connectors (Audio Jacks, USB-C, Micro SD, HDMI) (Qty: 4) — \$38.00

→ *Explanation: Interfaces external peripherals with internal logic, testing expandability and port access.*

- Microphone (Qty: 2) — \$22.00
→ *Explanation: Enables sound input testing and interface for voice and multimedia applications.*

- PCB Manufacturer (Qty: 1) — \$800.00
→ *Explanation: Used for creating custom printed circuit boards, necessary to verify real-world circuit integration and connections.*

- Copper Heat Pipes and Spreader (Qty: 1) — \$67.00
→ *Explanation: Enables thermal dissipation validation for critical power-intensive components, ensuring system stability.*

- Thermal Pads and Paste (Qty: 1) — \$50.00
→ *Explanation: Facilitates heat transfer between components and cooling systems for realistic stress tests.*

- Camera Module (Qty: 4) — \$42.00
→ *Explanation: Tests multimedia functionality and interface with image signal processing on the SoC.*

- Miscellaneous Components (Qty: 1) — \$210.00
→ *Explanation: Includes resistors, capacitors, and supporting elements necessary for various hardware subsystems.*

- ESP-32 Modules (Qty: 10) — \$80.00
→ *Explanation: Used for auxiliary processing, wireless comms, and prototyping of control interfaces.*

Milestone 2: Tooling and Frame Construction — Total: \$2,945.00

- This milestone supports system-wide development by funding essential fabrication and diagnostic equipment, enabling us to assemble, modify, and test components as designs evolve. Tools like heat guns, hot plates, cable testers, and clamp meters are critical to soldering and verifying high-density interconnects in our boards.

This phase also includes materials for the phone's first physical enclosure, machined from aluminum, which allows us to test thermals, fit, and user ergonomics. A second PCB manufacturing stipend is allocated for revised boards based on feedback from

Milestone 1. Additionally, we've included a buffer for unexpected equipment or component needs to prevent timeline disruptions during rapid prototyping and testing.

- PCB Manufacturer (Stipend) — \$800.00
- Heat Gun (Qty: 1) — \$165.00

→ *Explanation: Essential for safely reworking and assembling surface-mount components during board fabrication.*

- USB/Cable Tester (Qty: 1) — \$120.00

→ *Explanation: Tests integrity and throughput of USB and data ports, validating connectivity across modules.*

- Multimeter and Clamp Meter Kit (Qty: 1) — \$420.00

→ *Explanation: Provides voltage, resistance, and current measurement across subsystems for power and signal diagnostics.*

- Hot Plate (Qty: 1) — \$90.00

→ *Explanation: Used for thermal soldering tasks and batch surface-mount work, critical during rapid iteration.*

- Additional Equipment/Components — \$750.00

→ *Explanation: Buffer for unplanned purchases, revisions, or expansion modules across all teams.*

- Aluminum Phone Frame (Raw Aluminum + Finishing) (Qty: 1) — \$600.00

→ *Explanation: Physical housing to test form factor, thermal behavior, and user ergonomics for prototype deployment.*

Milestone 3: Final Integration + Team Compensation — Total: \$1,872.50

- The final phase supports the full system integration and team coordination needed to deliver a working prototype. A stipend is allocated for the Project Manager who has led project coordination, risk mitigation, documentation, and cross-team synchronization throughout the project. An additional flexible stipend is budgeted for key contributors who go above and beyond during integration, validation, and final handoff.
- Project Manager Stipend — \$1,872.50

→ *Explanation: Funds the coordination and integration responsibilities of the PM, who ensures progress across all deliverables.*

- Team Member Stipend (Tentative) — TBD

→ *Explanation: Reserved for contributors going beyond baseline effort, especially in critical integration phases.*

- Total Project Budget: \$7,250.00

References

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